# Benchmark for Crack Re-Identification in Ticket Management Systems

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***Abstract*—In a ticket-based maintenance system, each new photograph of a crack should be assigned to the correct ticket, but if the same crack is photographed from a different angle it may be treated as a new defect, breaking its historical record and leading to redundant inspection work. We introduce CRACKID, a dataset of 140 photographs from 17 walls containing 212 distinct crack identities, and use it to assess ten zero-shot matchers under three scenarios: retrieval, same-defect decisions, and open-set identification with rejection. Ranking quality varies considerably (Rank-1 0.352–0.654), yet validation-calibrated same-defect decisions stay close to the constant baseline ($F_1$ 0.293–0.351 vs. 0.305). Context and crack-erasure experiments suggest that wall-context cues matter more than crack appearance. A geometric reference that aligns images using crack-free wall texture and compares the projected crack masks achieves a detection and identification rate of 0.768 at a false-alarm rate of 0.1 on the 48% of pairs that can be aligned, compared with 0.429 for the best crop-based method scored on all valid pairs. These point estimates are from a small wall-clustered sample and cover only single-visit correspondence. They suggest that good retrieval alone does not guarantee reliable crack association, and that local visual context and geometric alignment may both be needed. CRACKID provides a foundation for developing more reliable visual association methods, with cross-visit evaluation and ticket-level duplicate reduction as future work.**

***Index Terms*—Benchmark, Building Maintenance, Crack Re-Identification, Geometric Alignment, Homography**

## I. INTRODUCTION

Ticket management systems record physical defects for inspection, prioritization, repair, and follow-up. Ideally, each ticket holds one incident's history, but users often photograph the same crack more than once, creating duplicate tickets that fragment this history and cause redundant work. Each new observation must therefore be linked to an existing ticket or open one for unrecorded damage.

This is a visual re-identification (re-ID) problem: the same crack must be recognized despite changes in viewpoint, distance, field of view, illumination, and visibility, and despite growth by extension, widening, or branching, while new damage must be distinguished from known damage. Errors in either direction lead to duplicate tickets or misattributed history. Crack detection and segmentation work focuses on single images. CrackSeg9k [1] and OmniCrack30k [2] contain no cross-observation identities, so observations cannot be associated with tickets. Whether person and object re-ID [3], [4] transfers to cracks is unclear: distinct defects may look alike while one defect can vary greatly.

This work introduces CRACKID to evaluate whether photographs depict the same wall crack: 140 photographs of 17 walls, each photographed during a single visit. Real photographs capture within-visit acquisition changes; photometric and growth changes are probed only synthetically. Cross-visit evaluation and ticket-level duplicate reduction are left for future work. All accuracy figures are point estimates from a small wall-clustered sample.

We assess ten zero-shot matchers using CRACKID. The protocol separates retrieval from the same-defect decision and includes open-set identification with rejection: the top-ranked match is accepted only above a validation-calibrated threshold, otherwise the query is rejected as a new ticket. Rejection is necessary because a high ranking alone is insufficient evidence for a link. To test whether matchers recognize the crack or only wall context, we use crack-erased controls and a geometric reference that aligns wall images and compares projected crack masks.

Three research questions guide the study:

- **RQ1.** Can the evaluated methods associate observations of the same defect under varying image acquisition conditions?
- **RQ2.** How do wall context and geometric alignment contribute to defect re-identification?
- **RQ3.** How do two selected approaches perform under synthetic photometric and growth changes?

RQ1 and RQ2 use real single-visit photographs; RQ3 is exploratory synthetic work.

The contributions are as follows: (i) the cross-observation identities of CRACKID, together with their annotation provenance (Section III-B); (ii) a benchmark for retrieval, same-defect decisions, and open-set identification, addressing RQ1 (Sections IV, VI-B, and VII); (iii) context controls, a geometric reference, and coverage analysis, addressing RQ2 (Sections III, V, VI-C, and VII-B); and (iv) a synthetic revisit probe for RQ3 (Section VII-C).

## II. RELATED WORK

Most crack analysis detects or segments cracks in individual images [5]–[7]. Consolidated datasets [1], [2] keep this image-level formulation without cross-observation identities. Multi-view pipelines fuse images from one inspection into surface models or segmentations [8] but do not represent persistent identities across observations. Structural health monitoring tracks known cracks over time with fixed cameras, designated

locations [9]–[11], or guided setups [12], assigning identities when monitoring is set up; our task is open-set association.

Our setting combines object re-ID [4] with open-set identification [13]. The methods assessed in this study are SIFT [14] and ORB [15], learned correspondences such as SuperGlue [16] and LoFTR [17], and crop embeddings [3], [18]–[22]; all are applied zero-shot, without fine-tuning on CRACKID (Section VI-A). LightGlue [23] and NetVLAD [24] are left as future baselines (Section VI-A). Vehicle-damage re-ID across claims [25] follows our open-set protocol, whereas other studies focus on detection and severity rather than association [26]–[28]. Aligning first and then reasoning about presence or absence is the approach of change detection [29], [30], which relies on planar registration [31]–[33]; we add a fallback based on enhanced correlation coefficient (ECC) maximization [34] (Section V-B). Our key distinction is that alignment and matching targets must be disjoint: the homography is estimated from the wall, not the crack.

## III. DATA ACQUISITION

### A. Acquisition and composition

The CRACKID dataset [35] consists of 140 photographs of 17 exterior walls taken at a single site over four half-days using two consumer handheld devices (Fig. 1). The walls are either stippled render (textured) or smooth painted plaster, and the only visible defects are cracks. Six walls are used for validation (01–06; 66 photographs, 66 identities) and eleven for testing (07–17; 74 photographs, 146 identities). Only identities in at least two photographs are *matchable* (18 validation, 19 test); their queries are the known (answerable) ones. The *frozen snapshot*, used for all results, comprises walls 01–17 of Version 1 of the released dataset [35]. The *live dataset* is the full current release, which also contains walls 18 and 19, added after the snapshot was frozen and excluded from results. Each wall was photographed in a continuous walk-around (median 3 s between frames, max 81 s), so the data include within-visit viewpoint and scale changes. This is more favorable than multi-visit matching, though not a formal upper bound. The frozen snapshot has 439 known and 74 unknown validation queries, and 280 known and 258 unknown test queries. Positive query–gallery pairs number 26,012 (validation) and 5,722 (test); valid pairs (both instances annotated) number 89,118 and 31,822. Of 301 same-wall test image pairs, 255 contain valid query–gallery pairs and 97 register (32% of the 301); 15,328 of the 31,822 valid test pairs (48%) belong to registered image pairs; we call these query–gallery pairs *registered pairs*.

### B. Annotation

The labeling pipeline propagates each click frame-to-frame with ORB homographies, then merges fragments into physical-crack identities; this differs from the ORB matcher benchmarked in Table I. Merging is the main source of label risk, since an incorrect merge can turn a hard negative pair into a falsely labeled positive. In the frozen snapshot, 95 of the 212 identities (45%) result from merges, including 70 of the 146 test identities (48%), with median bridged gaps of 228 px (test) and 286 px (validation). One annotator reviewed all clicks, including 1,023 unclaimed points (not assigned to any identity): 45 were not cracks, 62 were placed off the crack on the surrounding wall, and 916 were cracks missed by the mask (Section V). Inter-annotator agreement was not assessed. This review reduces but does not eliminate merge risk. The live dataset must not be combined with the frozen figures without re-running the benchmark.

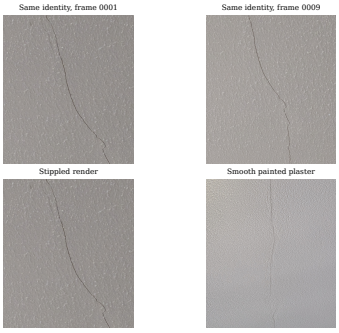



Fig. 1. CRACKID examples. Top: one identity across non-adjacent frames. Bottom: stippled render versus smooth plaster. Labels were propagated via homographies and reviewed by a single annotator.

## IV. BENCHMARK PROTOCOL

The benchmark determines whether a new crack observation matches a recorded defect. Annotated identities stand in for permanent records, so scores indicate association accuracy, not ticket-level duplicate reduction.

### A. Task formulation and candidate records

Every query and its gallery come from one wall, and an identity links all observations of the same crack (Section III-B). Cross-wall search is disallowed; the wall identity is supplied to every method. A query is a crack instance in one photograph; its gallery contains instances from the other photographs of the same wall. A query is known if its gallery contains another instance of the same identity, and unknown otherwise (a genuinely new defect). In closed-set retrieval, the true match is always in the gallery, so candidates are only ranked. Open-set identification mirrors ticket association: link the query to the best existing ticket or reject it and open a new one. Method scores drive ranking, thresholded decisions, and identification with rejection. Pairs are the unit for same-defect decisions; queries are the unit for retrieval and for open-set decisions; image pairs are the unit for registration and assignment. The Hungarian one-to-one constraint is applied only within each image pair.

### B. Test conditions, decisions, and metrics

The primary test uses a walk-around of each wall, during which crack morphology does not change. The only variations are viewpoint, distance, scale, lighting, and visibility; the galleries also contain same-wall distractors, and some

queries are unknown. The validation and test walls are entirely separate. All methods are applied off the shelf to the shared masks (Section V), without fine-tuning on CRACKID (Section VI-A). Thresholds are calibrated on the validation walls and applied unchanged to the test walls; only within-visit correspondence is considered. A separate synthetic probe isolates growth (condition A: one crack tip extended) from photometric change (condition B: blur, noise, and contrast changes with unchanged structure), comparing OSNet with context (OSNet+ctx, Section VI-A) against HYBRID-SKEL (Chamfer tolerance $\tau = 8.0$ px). Unlike REG+CHAMFER (Section V), which performs registration-gated Chamfer comparison only, HYBRID-SKEL blends alignment-based comparison with a crack-skeleton score when homography is available and falls back to skeleton matching otherwise. The probe is preliminary.

*Decisions.* Pairs scoring above the validation-calibrated threshold are accepted. For open-set identification, only the highest-ranked entry is accepted, and only above the threshold; otherwise the query is rejected as a new ticket. Within each image pair, a Hungarian one-to-one assignment is computed and matches below the threshold are discarded. Unscorable pairs (for example, unregistered) are retained: they are predicted negative for $F_1$ and ranked last; a registration failure is absence of evidence, not evidence of a new defect.

*Metrics.* Closed-set ranking uses Rank-1 and mean average precision (mAP); thresholded decisions use pairwise $F_1$ at the oracle threshold ($F_1^*$; the test-set $F_1$ maximizer) and at the validation-calibrated threshold ($F_1^{@v}$); assignment uses $F_1^a = 2\text{TP}/(2\text{TP} + \text{FP} + \text{FN})$, where TP, FP, and FN denote true positives, false positives, and false negatives. Open-set identification uses the detection and identification rate (DIR) at a false-alarm rate (FAR) of 0.1 (the fraction of known queries correctly linked when at most 10% of unknown queries are wrongly linked), plus coverage (the fraction of valid query–gallery pairs scored). Reference levels are random ranking and the constant baseline, which labels every pair as positive and achieves $F_1 = 2p/(1 + p)$, where $p$ is positive prevalence. This yields $F_1 = 0.305$ on the full test set ($p = 0.180$), $0.544$ on the geometric subset ($p = 0.373$; $48\%$ of valid pairs), and $0.452$ on validation ($p = 0.292$). Test-set reference levels are in Table I. Because only 11 walls and 19 matchable identities are available, we report point estimates only (Section VIII).

## V. A GEOMETRIC REFERENCE

The geometric reference tests whether wall crack locations drive association: it aligns wall surfaces to a common frame and compares projected crack masks. We call this registration-gated Chamfer comparison REG+CHAMFER (query–gallery pairs whose images fail to register are not scored) and its registration-only variant, which assigns a constant score to registered pairs, REG-ONLY. The geometric reference assumes a roughly planar wall and uses no CRACKID training. Shared masks come from a segmentation model trained on public crack data only (no CRACKID wall in training). Crack instances use a minimum area of 200 pixels and a 5-pixel closing kernel, so all methods receive consistent inputs. Masks are taken as given, and their error is not measured.

### A. Registration using wall texture

For images $I_a$ and $I_b$, a homography $H$ ($x' \sim Hx$) is estimated from surrounding wall texture with cracks dilated and excluded, so alignment is driven by surface features independently of the tested identity. This independence is conditional on the masks and on the cascade stage, not guaranteed per pair. Full detector settings (SIFT with contrast-limited adaptive histogram equalization (CLAHE) [36], ratio test, robust estimation) are archived with the code.

### B. Registration cascade and validity checks

Textured render generally provides background features, whereas smooth plaster does not. The cascade has three stages: masked SIFT; unmasked SIFT if too few keypoints remain (flagged, since crack features can then influence alignment); and photometric ECC [34]. Candidate solutions are checked for plausibility: the quadrilateral must not be mirrored or folded, and the scale must lie within $[1/16, 16]$; no minimum inlier count is imposed. We record the successful stage per pair; its inlier root-mean-square reprojection error (RMSE) sets the comparison tolerance. Absence of registration means lack of evidence, not that the cracks differ.

### C. Crack projection and geometric comparison

Accepted registrations project instances from $I_a$ into $I_b$ and compare them using the symmetric mean Chamfer distance [37], defined in (1). Let $a$ and $b$ be crack point sets and $d(p, b) = \min_{q \in b} \|p - q\|_2$:

$$D(a, b) = \frac{1}{2}\left[\frac{1}{|a|}\sum_{p \in a} d(p, b) + \frac{1}{|b|}\sum_{q \in b} d(q, a)\right]. \quad (1)$$

Unlike intersection over union (IoU), which is overly sensitive to small misalignments of narrow cracks, Chamfer tolerates them. Candidates whose Chamfer distance exceeds the validation-calibrated threshold are rejected. A scale-free alternative is Chamfer coverage, the fraction of crack points within the per-pair RMSE (reprojection error) of the other mask; tolerance is clamped to $[2, 16]$ px, default 6 px.

### D. Correspondence decisions and limits of spatial evidence

For each image pair, one-to-one matching uses the Hungarian method [38] at the validation-calibrated threshold; unregistered pairs score $-\infty$ (Section IV). Alignment onto a crack-free region is evidence of absence; missed detections, out-of-frame areas, and registration failures leave the question unresolved. The reference assumes that damage does not change; its performance may depend on the wall surface and on label provenance (Section III-B).

TABLE I

MAIN RESULT, TEST (11 WALLS, $n_Q$=280 KNOWN, 258 UNKNOWN). POINT ESTIMATES; THE EFFECTIVE SAMPLE IS 11 WALLS. METRICS: R@1 RANK-1 ACCURACY; MAP MEAN AVERAGE PRECISION; DIR DETECTION AND IDENTIFICATION RATE AT FAR 0.1 (FRACTION OF KNOWN QUERIES CORRECTLY IDENTIFIED WHEN $\leq$10% OF UNKNOWN QUERIES ARE WRONGLY LINKED); $F_1^*$ ORACLE-THRESHOLD $F_1$; $\Delta F_1^*$ EXCESS OVER THE CONSTANT BASELINE (0.305 CROP-BASED, 0.544 GEOMETRIC); $F_1^{@v}$ VALIDATION-CALIBRATED $F_1$; COVERAGE: FRACTION OF VALID QUERY–GALLERY PAIRS SCORED. BOLD: BEST CROP-BASED METHOD PER COLUMN. THE GEOMETRIC REFERENCE SCORES 48% OF VALID PAIRS AND RANKS THE REST LAST, SO THE DIR VALUES 0.768 AND 0.429 USE DIFFERENT CANDIDATE POOLS.

| Method | Family | R@1 | mAP | DIR @FAR = 0.1 | $F_1^*$ | $\Delta F_1^*$ | $F_1^{@v}$ | coverage |
|---|---|---|---|---|---|---|---|---|
| *Chance (random ranking)* | | | | | | | | |
| Chance | reference | 0.336 | 0.385 | 0.038 | 0.305 | 0.000 | 0.305 | 1.00 |
| *Reference level (same 0.48 pool as geometry)* | | | | | | | | |
| REG-ONLY | geometry | 0.515 | 0.533 | 0.000 | 0.544 | 0.000 | 0.544 | 0.48 |
| *Crop scope* | | | | | | | | |
| ORB [15] | keypoint | 0.352 | 0.364 | 0.018 | 0.305 | 0.000 | 0.305 | 1.00 |
| SIFT [14] | keypoint | 0.432 | 0.417 | 0.036 | 0.313 | 0.008 | 0.305 | 1.00 |
| SuperGlue [16] | learned | 0.562 | 0.455 | **0.429** | 0.353 | 0.048 | 0.305 | 1.00 |
| LoFTR [17] | learned | 0.549 | 0.440 | 0.336 | **0.359** | **0.054** | **0.351** | 1.00 |
| YOLOv8 backbone [22] | embedding | 0.561 | 0.453 | 0.157 | 0.357 | 0.052 | 0.308 | 1.00 |
| ViT-B/16 [18] | embedding | **0.654** | 0.511 | 0.286 | 0.331 | 0.026 | 0.317 | 1.00 |
| DeiT-S [19] | embedding | 0.621 | 0.497 | 0.250 | 0.305 | 0.000 | 0.293 | 1.00 |
| CLIP ViT-B/32 [20] | embedding | 0.604 | 0.471 | 0.143 | 0.307 | 0.002 | 0.306 | 1.00 |
| DINOv2 [21] | embedding | 0.607 | 0.484 | 0.307 | 0.305 | 0.000 | 0.297 | 1.00 |
| OSNet [3] | re-ID | **0.654** | **0.520** | 0.168 | 0.309 | 0.004 | 0.309 | 1.00 |
| *Full-photograph scope* | | | | | | | | |
| REG+CHAMFER (reference) | geometry | 0.952 | 0.886 | 0.768 | 0.656 | 0.112 | 0.582 | 0.48 |

## VI. OFF-THE-SHELF APPEARANCE BASELINES VERSUS FULL-PHOTOGRAPH REFERENCE

### A. Setup

Ten matchers operate on crack-centered boxes taken from the shared masks. SIFT, ORB, SuperGlue, and LoFTR are scored by RANSAC inlier counts, with indoor weights for SuperGlue and LoFTR; the five embedding models (listed in Table I) and OSNet are compared by cosine similarity of L2-normalized crop embeddings. None is fine-tuned on CRACKID. For YOLOv8 [22], we use layer-9 features of `yolov8n.pt`; model sizes and checkpoints are in the code. Context views extend the box by one width and height on each side (three times as wide and tall); OSNet with this crop is OSNet+ctx. In erased views, the crack is dilated by 5 px and inpainted with Telea's method [39]; artifacts are possible, and the treatment of other fragments and neighboring cracks is not described. Context and erasure views were tested only for OSNet and CLIP; no crack-only or wall-only views were evaluated. The appearance baselines operate on tight crops, whereas the geometric reference uses full photographs. We do not evaluate full-photograph descriptors (DINOv2 on whole photographs, NetVLAD [24]), LightGlue [23], or matcher-fed Chamfer (Chamfer after registering with a learned matcher). Runtimes are in the code repository; sessions were not load-controlled, so they do not rank methods.

### B. Ranking quality versus thresholded decisions

Ranking quality varies widely, while thresholded decisions stay near the constant baseline (Table I). Rank-1 ranges from 0.352 to 0.654, mAP reaches 0.520, and crop DIR reaches 0.429. The oracle-threshold $F_1$ ranges from 0.305 to 0.359 and the validation-calibrated $F_1^{@v}$ from 0.293 to 0.351. Relative to the 0.305 constant baseline, the validation-calibrated excess is between $-0.012$ and $+0.012$ for every method except LoFTR ($+0.046$); ORB, SIFT, and SuperGlue accept all pairs at the validation-calibrated threshold ($F_1 = 0.305$). AUROC, average precision (AP), DIR–FAR curves, and wall-clustered confidence intervals require archived score matrices, so we do not report them; leave-one-wall-out evaluation over all 17 walls would be preferable to the 6/11 split (Section VIII). The constant-baseline $F_1$ falls from 0.452 (validation) to 0.305 (test) because prevalence shifts from 0.292 to 0.180, so absolute $F_1$ values are not comparable across splits. With 19 matchable identities on 11 walls, we do not claim a fine-grained ranking or separability from chance.

### C. Crack appearance versus wall context

For OSNet, context raises DIR from 0.168 to 0.404 at FAR 0.1; erasing the crack leaves DIR at 0.393 (Rank-1 0.711 vs. 0.775 with context), suggesting that location-specific context carries much of the signal. For CLIP, DIR rises from 0.143 to 0.311 with context and to 0.286 after erasure, showing a similar pattern. These differences are point estimates without confidence intervals. The context box mostly adds neighboring cracks, possibly sibling fragments (45% of identities result from merges, Section III-B); the signal likely comes from context around cracks rather than wall texture alone. Crack-only and wall-only views on more backbones with confidence intervals would sharpen this. Across these controls, $F_1^*$ stays within 0.307–0.361.

## VII. RESULTS AND DISCUSSION

### A. Main comparison

The geometric reference scores only registered pairs (48% of valid query–gallery pairs, containing all 5,722 positive

pairs), whereas crop-based methods score all valid pairs; the two result sets use different candidate pools. With REG-ONLY (a constant on registered pairs, $-\infty$ otherwise), Rank-1 is 0.515, mAP 0.533, and DIR 0.000 at FAR 0.1 (zero because all registered pairs score equally). Adding mask comparison improves Rank-1 and mAP by 0.437 and 0.353, respectively, and raises DIR to 0.768 (Table I). The highest crop-based DIR values are 0.429 (SuperGlue) and 0.404 (OSNet+ctx) (Section VI-C), against chance 0.038. No wall-clustered confidence intervals are reported. At the validation-calibrated threshold, REG+CHAMFER exceeds its constant baseline by only 0.038 (0.582 vs. 0.544), compared with 0.112 at the oracle threshold ($F_1^* = 0.656$). This excess is smaller than LoFTR's 0.046 over its 0.305 baseline, suggesting that the contrast seen in ranking and DIR overstates the difference in thresholded decisions. Registration gating may not help across visits; an appearance baseline restricted to registered pairs and the frame-gap baseline (ranking by the number of frames between photographs) remain to be evaluated.

### *B. Registration rate (Section III-A) and failure modes*

The registration gate is label-informative: it retains all positives but only 36.8% of negatives, which raises the prevalence from 0.180 to 0.373 and the constant-baseline $F_1$ from 0.305 to 0.544.

Registration failures depend on wall surface (Section V-B); their extent on smooth plaster is unquantified. All 280 known queries have a true match in an adjacent frame (median gap 1), motivating the frame-gap baseline.

We define sharpness as the variance of the Laplacian of the image at quarter resolution. A logistic regression on the test set suggests sharpness has a larger standardized association with registration success than frame gap ($|z| = 5.11$ vs. 2.78); with 11 clusters, however, the $p$-values overstate precision, and the sharpness threshold of 10, used to predict registration success, was chosen post hoc. Registration, ECC, and crack-exclusion ablations were run on the test set and are not reported here; geometry settings should be fixed on validation before any final comparison.

### *C. Synthetic revisit probe*

Ten photographs from five walls were altered by a small common warp under condition A or B (Section IV), with one random seed. Altered images were matched against real photographs. The warp is small (scale 1.08, rotation 6°, out-of-plane tilt 4°), whereas real test pairs reach 75° out-of-plane tilt at the 90th percentile (procedure in the code). Table II compares HYBRID-SKEL and OSNet+ctx. For ranking and open-set identification (A/B), OSNet+ctx outperforms HYBRID-SKEL (Rank-1 0.826/0.671 vs. 0.275/0.274; DIR 0.739/0.663 vs. 0.343/0.342; $F_1^{@v}$ 0.601/0.594 vs. 0.569/0.574); assignment $F_1^a$ reverses narrowly (0.631/0.609 vs. 0.646/0.640), and no pair was scored by homography alone, so the reversal does not come from a homography-only path. For RQ3, this single-seed probe provides no evidence of real multi-visit robustness, which remains untested (Section VIII).

TABLE II

SYNTHETIC REVISIT PROBE: HYBRID-SKEL VERSUS OSNET+CTX FOR A (GROWTH) AND B (PHOTOMETRIC CHANGE), ONE SEED, FIVE WALLS. $n_Q$ NUMBER OF QUERIES; R@1 RANK-1 ACCURACY; MAP MEAN AVERAGE PRECISION; DIR DETECTION AND IDENTIFICATION RATE AT FAR 0.1; $F_1^{@v}$ VALIDATION-CALIBRATED $F_1$; $F_1^a$ ASSIGNMENT $F_1$. HYBRID-SKEL ADDS A CRACK-SKELETON SCORE TO ALIGNMENT-BASED COMPARISON; REG+CHAMFER USES REGISTRATION-GATED CHAMFER ONLY. BOLD: BEST METHOD PER COLUMN AND CONDITION.

| Condition | Method | $n_Q$ | R@1 | mAP | DIR @FAR$_{0.1}$ | $F_1^{@v}$ | $F_1^a$ |
|---|---|---|---|---|---|---|---|
| A (growth) | HYBRID-SKEL | 69 | 0.275 | 0.401 | 0.343 | 0.569 | 0.646 |
| | OSNet+ctx | 69 | **0.826** | **0.647** | **0.739** | 0.601 | 0.631 |
| B (photometric) | HYBRID-SKEL | 82 | 0.274 | 0.403 | 0.342 | 0.574 | 0.640 |
| | OSNet+ctx | 82 | **0.671** | **0.570** | **0.663** | 0.594 | 0.609 |

## VIII. LIMITATIONS AND FUTURE WORK

This study presents the first quantitative evaluation of re-identifying a crack in a single visit using a geometric reference. The results are limited by the small number of clustered walls, single-annotator labels, and partial registration coverage. Each item pairs a finding with a verification step.

1) *Scale to more walls and visits.* Collect data from more walls and surface types and annotations from additional annotators, tightening wall-clustered confidence intervals and making rankings more reliable; report leave-one-wall-out results over all 17 walls instead of the 6/11 split. Most importantly, add a second session with different distances and lighting for a genuine cross-visit benchmark.
2) *Add provenance-free annotation.* Labels come from a single annotator (Section III-B) and were built by homography propagation, so the geometric advantage may depend on provenance. To test whether it persists independently of label provenance, create labels that do not rely on homography propagation and release per-pair score matrices for wall-clustered confidence intervals and threshold-free metrics (AUROC, AP).
3) *Close the coverage gap.* The geometric reference covers 48% of valid query–gallery pairs (15,328 of 31,822), from the 97 of 301 same-wall image pairs (32%) that register. It tends to fail on smooth plaster, with extent unquantified. Surface-adaptive registration (learned local features, direct photometric alignment, or sparse depth from consumer devices) would allow a more thorough evaluation and better separate geometric evidence from homography-based provenance.
4) *Learn from geometric supervision.* The geometric reference is untrained and not deployed. Location cues could be obtained without per-pair registration at inference time via registration-derived pseudo-labels or fine-tuning a foundation model on homography correspondences.
5) *Assess the ticket lifecycle.* Define task metrics such as duplicate rate and missed-new-defect rate, and test segmentation, re-identification, and joint ticket assignment in a closed-loop simulation.

Together, these steps would turn the within-visit reference into a multi-visit component usable in operational ticket systems.

## IX. Conclusion

We present CRACKID, which consists of 140 photographs from 17 walls, along with a protocol for retrieval, same-defect decisions, and open-set identification with rejection.

The main finding is negative. For RQ1, ten zero-shot crop matchers exceed chance Rank-1 in point estimates, but with a validation-calibrated threshold their $F_1$ stays near the constant baseline. For RQ2, wall context appears to contribute more than crack appearance, and the geometric reference improves open-set identification only on the 48% of alignable valid pairs, with results that may depend on annotation provenance. For RQ3, the small synthetic revisit probe provides no evidence of multi-visit robustness; OSNet+ctx leads on retrieval and open-set identification, whereas HYBRID-SKEL is marginally better on assignment $F_1$.

The study focuses on single-visit photographs: the figures are point estimates derived from 19 matchable test identities, grouped by wall and evaluated against labels from a single annotator. The next step is a second acquisition session to build a cross-visit benchmark, followed by ticket-level evaluation.

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